

Analisa Intelijen Energi Indonesia Kedepan

Final - Strategic Intelligence Report 2025

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Ringkasan Eksekutif / Executive Summary

Laporan ini menyajikan kajian strategis komprehensif tentang posisi energi Indonesia dalam konteks geopolitik dan ekonomi global hingga 2050. Menggunakan pendekatan intelijen energi (energy intelligence), dokumen ini merangkum tren historis, sinyal peringatan dini, skenario strategis, serta rekomendasi kebijakan untuk memperkuat ketahanan dan kedaulatan energi nasional.

Intisari temuan: produksi minyak domestik mengalami penurunan struktural sementara produksi gas menunjukkan stabilitas relatif; investasi hulu berfluktuasi namun ada sinyal pemulihan, dan energi terbarukan menunjukkan percepatan penambahan kapasitas. Di tingkat geopolitik, peran Indonesia semakin krusial dalam rantai pasok LNG Asia Tenggara. Risiko utama meliputi volatilitas harga komoditas, keterlambatan investasi RE, dan ancaman keamanan maritim serta siber.

This report provides a comprehensive strategic assessment of Indonesia's energy position within the global geopolitical and economic context to 2050. Applying an energy intelligence approach, the document synthesizes historical trends, early-warning indicators, strategic scenarios, and policy recommendations to strengthen national energy resilience and sovereignty.

Key takeaways: domestic oil production faces structural decline while gas production remains relatively stable; upstream investment fluctuates but shows recovery signs, and renewable energy capacity additions are accelerating. Geopolitically, Indonesia's role in the Southeast Asian LNG supply chain is increasingly pivotal. Primary risks include commodity price volatility, delayed RE investments, and maritime and cyber security threats.

Daftar Isi / Table of Contents

DAFTAR ISI TERINTEGRASI: (Detail bab dan subbab pada halaman berikutnya)

I. Pendahuluan Strategis / I. Strategic Introduction

Latar belakang strategis: Energi sebagai pilar kebijakan nasional dan instrumen geopolitik yang memengaruhi stabilitas ekonomi dan pertahanan.

Strategic background: Energy as a pillar of national policy and a geopolitical instrument affecting economic stability and defense.

Signifikansi kajian: Menilai bagaimana perubahan iklim, transisi energi global, serta persaingan geopolitik memengaruhi pilihan kebijakan Indonesia.

Study significance: Assessing how climate change, global energy transition, and geopolitical competition affect Indonesia's policy choices.

Metodologi: Pendekatan OSINT, analitik ekonomi, pemodelan skenario, dan integrasi data spasial untuk memahami risiko dan peluang.

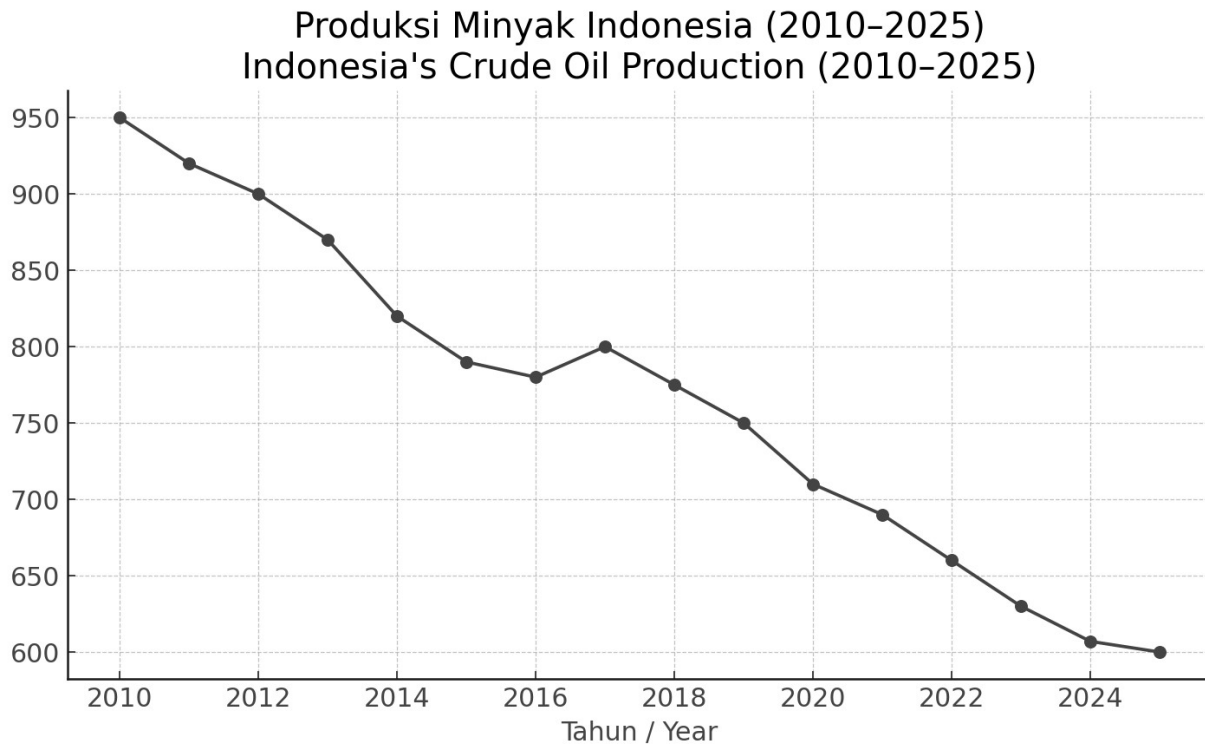
Methodology: OSINT approach, economic analytics, scenario modeling, and spatial data integration to understand risks and opportunities.

Kerangka konsep: Menggabungkan PESTEL, SWOT, dan threat-opportunity grid khusus energi untuk intelijen tindakan.

Conceptual framework: Combining PESTEL, SWOT, and a threat-opportunity grid specialized for actionable energy

intelligence.

Gambar. Produksi Minyak Indonesia (2010–2025) / Figure. Indonesia's Crude Oil Production (2010–2025)



II. Struktur & Ekosistem Energi Nasional / II. National Energy Ecosystem & Infrastructure

Struktur migas nasional: uraian pemerintahan, regulator, dan aktor industri (SKK Migas, ESDM, Pertamina, BUMN, swasta

internasional).

National oil & gas structure: overview of governance, regulators, and industry actors (SKK Migas, ESDM, Pertamina,

state-owned enterprises, international private firms).

Infrastruktur kritis: kilang, terminal LNG, pipa utama, serta titik kerawanan seperti choke points laut dan fasilitas

penyimpanan.

Critical infrastructure: refineries, LNG terminals, main pipelines, and vulnerability points such as maritime

chokepoints and storage facilities.

Peran hilirisasi & ekosistem manufaktur: nilai tambah dari pengolahan nikel, baterai, serta potensi integrated value

chain untuk energi bersih.

Downstreaming & manufacturing ecosystem: added value from nickel processing, batteries, and potential integrated value

chains for clean energy.

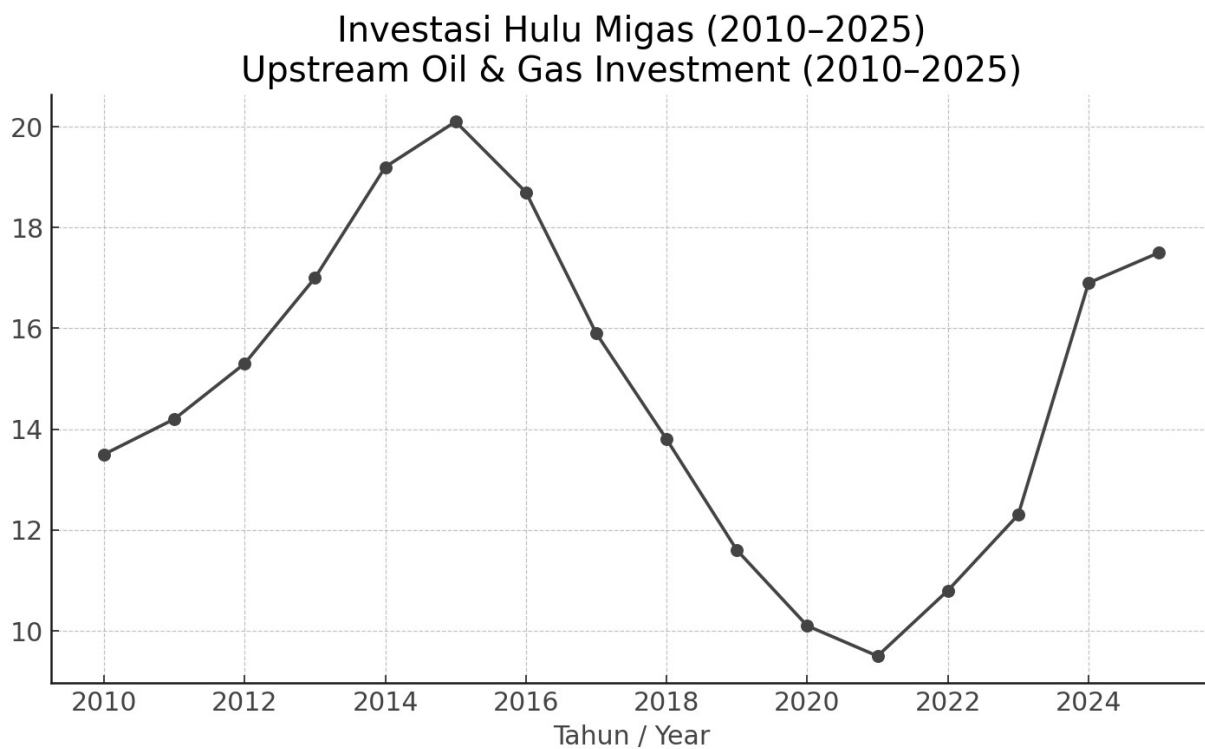
Rencana investasi & prioritas infrastruktur 2025–2035: fokus pada transmisi listrik, kapasitas penyimpanan, dan kilang

baru.

Investment plans & infrastructure priorities 2025–2035: focus on grid transmission, storage capacity, and new

refineries.

Gambar. Investasi Hulu Migas (2010–2025) / Figure. Upstream Oil & Gas Investment (2010–2025)



III. Intelijen Energi Global & Regional / III. Global & Regional Energy Intelligence

Jejaring intelijen energi global: aktor, tujuan, dan mekanisme pengumpulan data (satellite imagery, AIS, trade flows).

Global energy intelligence networks: actors, objectives, and data collection mechanisms (satellite imagery, AIS, trade flows).

Pengaruh geopolitik: kompetisi AS-China-Japan untuk akses LNG dan material kritis, serta dampaknya bagi Indonesia.

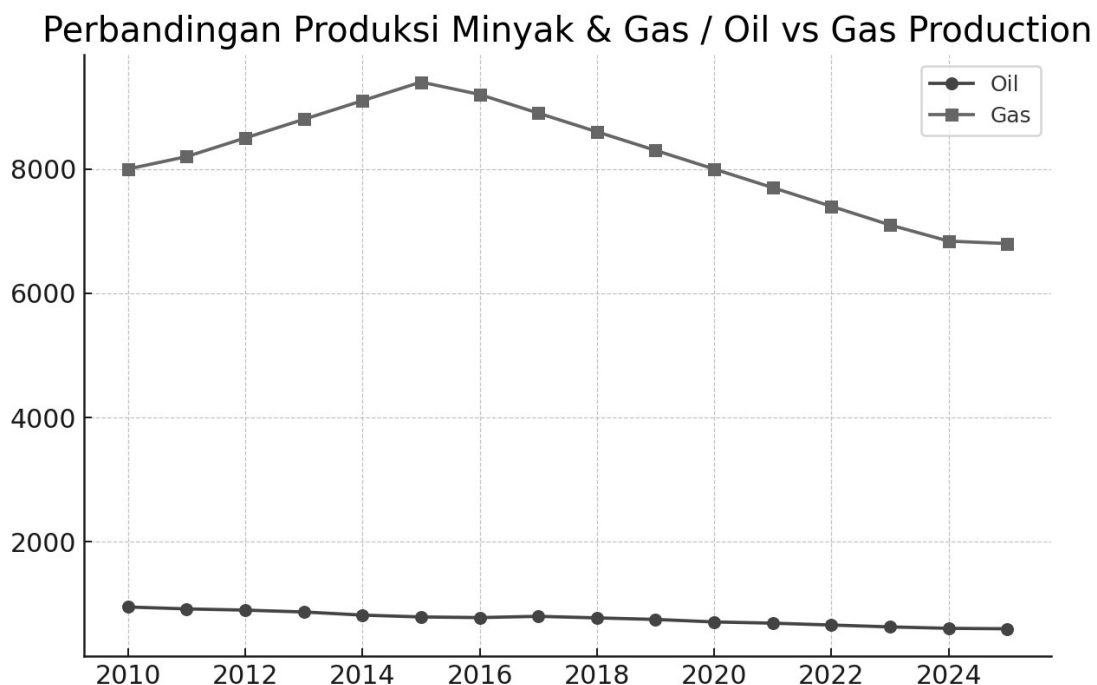
Geopolitical influence: US-China-Japan competition for LNG access and critical materials, and its implications for

Indonesia.

Peranan lembaga internasional: IEA, OPEC, JOGMEC, dan think tanks dalam membentuk norma serta akses modal proyek energi.

Role of international institutions: IEA, OPEC, JOGMEC, and think tanks in shaping norms and project financing access.

Gambar. Perbandingan Produksi Minyak & Gas / Figure. Oil vs Gas Production



IV. Keamanan & Pertahanan Energi Nasional / IV. National Energy Security & Defense

Konsep ketahanan energi nasional: definisi, metrik, dan hubungan dengan keamanan nasional serta stabilitas sosial.

National energy resilience concept: definition, metrics, and relationship with national security and social stability.

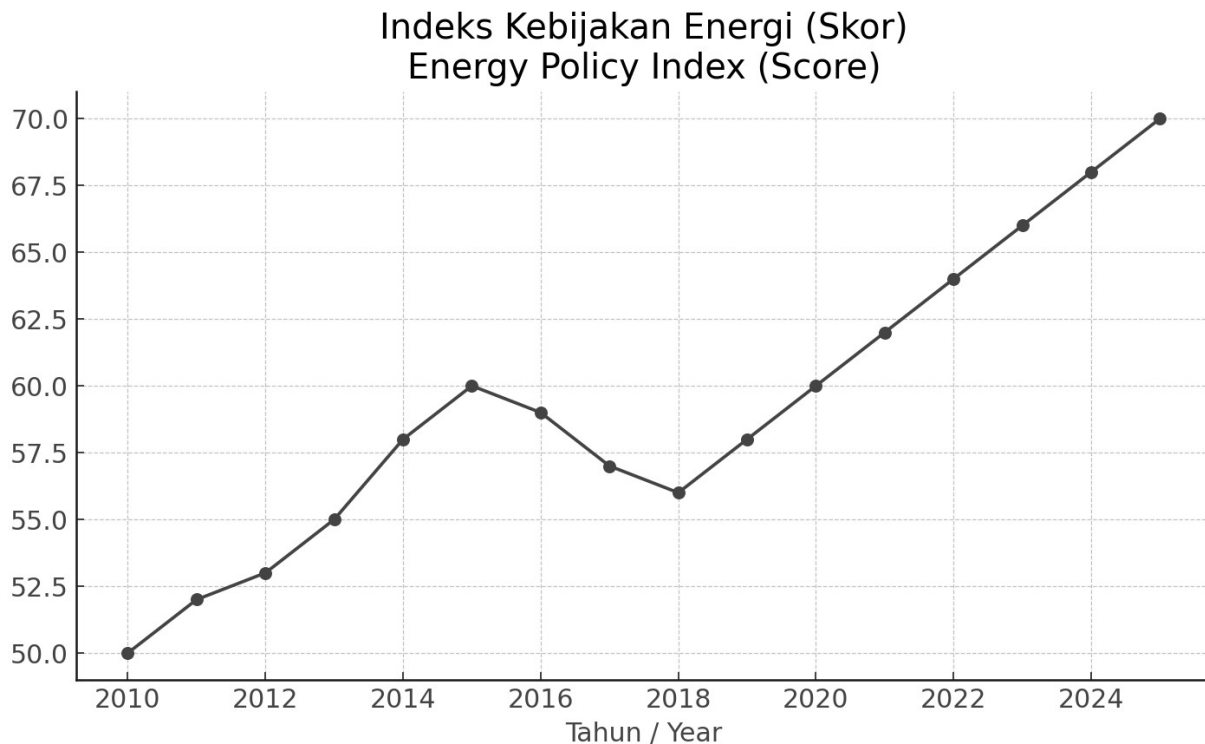
Ancaman maritim dan perbatasan: patroli, maritime domain awareness, dan ketegangan di Laut Natuna & Laut Cina Selatan.

Maritime and border threats: patrols, maritime domain awareness, and tensions in the Natuna Sea & South China Sea.

Keamanan siber energi: ancaman terhadap SCADA, ransomware pada operator, dan mitigasi melalui kerangka nasional.

Energy cybersecurity: threats to SCADA, ransomware on operators, and mitigation through a national framework.

Gambar. Jejaring Intelijen Energi Global / Figure. Global Energy Intelligence Network



V. Transformasi, Inovasi, dan Skenario 2050 / V. Transformation, Innovation, and 2050 Scenarios.

Transformasi energi: roadmap menuju 45% EBT pada 2050, pengembangan hydrogen, dan CCUS sebagai komponen transisi.

Energy transformation: roadmap to 45% RE by 2050, hydrogen development, and CCUS as transition components.

Inovasi & industrial policy: insentif R&D, transfer teknologi, dan peran kluster industri di Kalimantan dan Jawa.

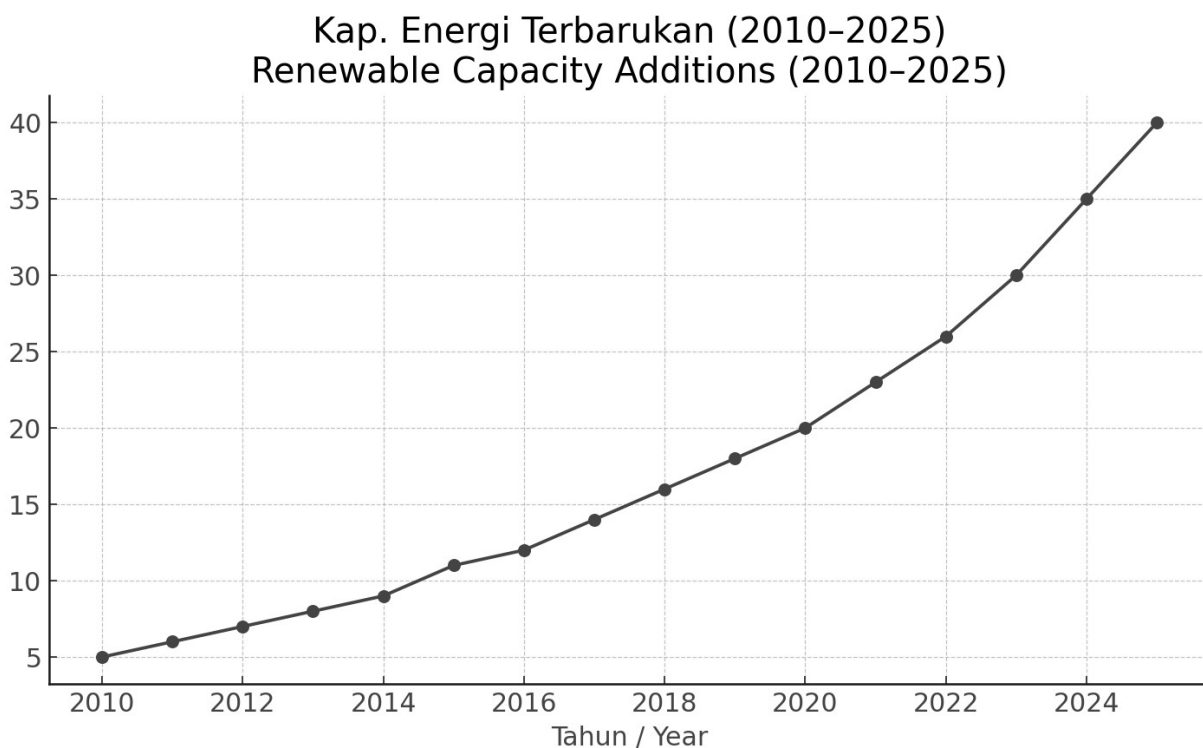
Innovation & industrial policy: R&D incentives, technology transfer, and role of industrial clusters in Kalimantan and

Java.

Skenario optimis dan alternatif: parameter utama, asumsi makro, dan implikasi kebijakan untuk tiap skenario.

Optimistic and alternative scenarios: key parameters, macro assumptions, and policy implications for each scenario.

Gambar. Kap. Energi Terbarukan (2010–2025) / Figure. Renewable Capacity Additions



VI. Kesimpulan Strategis & Rekomendasi / VI. Strategic Conclusions & Recommendations

Ringkasan temuan strategis: sinergi antara kebijakan ekonomi, keamanan, dan intelijen untuk mencapai kedaulatan energi.

Summary of strategic findings: synergy between economic, security, and intelligence policies to achieve energy

sovereignty.

Rekomendasi prioritas 2025–2050: konsolidasi data, reformasi fiskal, peningkatan infrastruktur, dan pengembangan SDM.

Priority recommendations 2025–2050: data consolidation, fiscal reform, infrastructure enhancement, and human capital

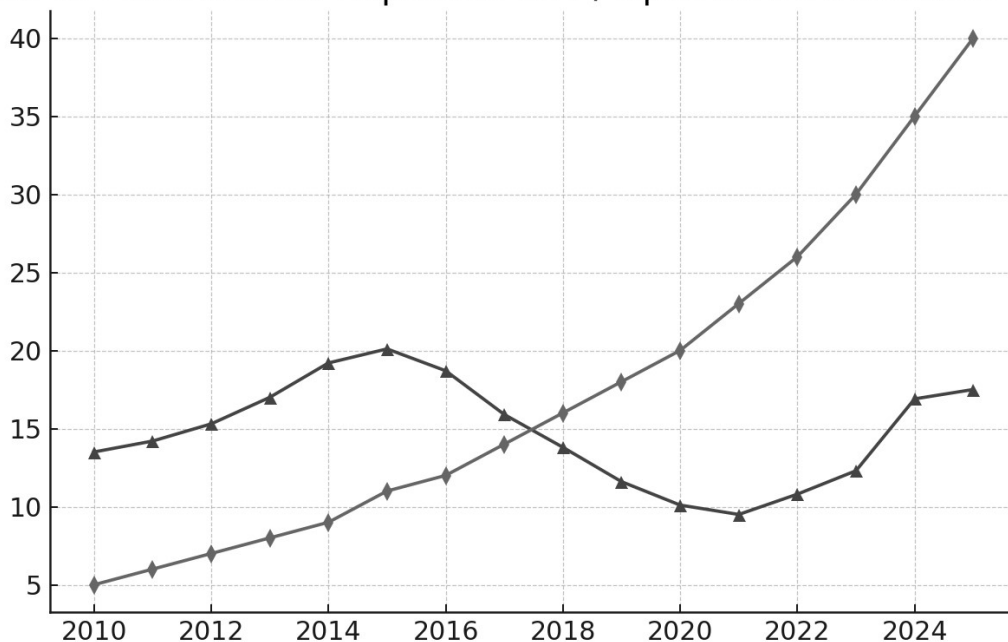
development.

Roadmap implementasi: langkah jangka pendek, menengah, dan panjang dengan indikator kinerja utama (KPIs).

Implementation roadmap: short-, medium-, and long-term steps with key performance indicators (KPIs).

Gambar. Investasi vs Penambahan Kapasitas EBT / Figure. Investment vs RE Additions

Investasi Hulu vs Penambahan Kapasitas EBT / Upstream Investment vs RE Additions



Lampiran / Appendices

Gambar Lampiran 1. Peta Skenematik / Appendix Map 1



Gambar Lampiran 2. Peta Skenematik / Appendix Map 2



Gambar Lampiran 3. Peta Skenematik / Appendix Map 3



Lampiran B: Tabel Data & Proyeksi / Annex B: Data Tables & Projections

Tabel 1: Data proyeksi dan analisis. Di sini disajikan proyeksi produksi, investasi, dan bauran

Daftar Referensi OSINT / References

- SKK Migas Annual Report
- BP Statistical Review of World Energy 2024
- International Energy Agency (IEA) Data
- U.S. EIA International Statistics
- World Bank DataBank
- IMF World Economic Outlook
- ASEAN Centre for Energy (ACE) reports
- Global Energy Monitor
- Reuters, FT, and regional press coverage

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Jakarta, 09 Oktober 2025